

AJAY VENKATESH

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Education

New York University, New York, NY

Sep 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Experience

- CampusMesh** Jan 2026 – Present
Software Engineer
– Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
– Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda, API Gateway, S3, and CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys, SSM Parameter Store**, and **least-privilege IAM**.
– Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
– Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access (**4+ roles**) using **OAuth, Midway SSO, and HELIS**.
– Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments**.
– Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
– Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; applied low-level design and design patterns, reduced backend execution time by **35%** and ensured reliability by unit and integration testing using **Mockito**.
- New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
Software Developer (On-Campus)
– Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.
– Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
– Developed an insurance analytics platform (**React, Node.js, REST APIs**); deployed **Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue.
– Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.
- PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
– Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
– Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
– Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
– Optimized data ingestion with **parallel processing** for **100K-row** from Excel; reduced execution time from **5 hours to 1 hour**.
– Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

Projects

- Advanced Expense Tracker (Java Desktop Application)** | *Java, JavaFX, JDBC, SQLite, JFreeChart*
– Architected a modular **Java application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
– Implemented concurrent transaction handling and real-time cost splitting using **multithreading** and synchronized blocks to ensure thread-safe database updates.
- Hospital Management System (Full-Stack Application)** | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*
– Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
– Fine-tuned a **Hugging Face transformer-based model** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.
- RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face, AGNews*
– Architected a **parameter-efficient fine-tuning (PEFT)** pipeline using a custom **Low-Rank Adaptation (LoRA)** implementation to optimize a **RoBERTa-base** model for news classification.
– Optimized model efficiency by freezing pretrained weights and injecting trainable rank-decomposition matrices, reducing trainable parameters to under **1M (934K)** while achieving **83.45%** accuracy on the private leaderboard.
– Conducted extensive hyperparameter sweeps for rank (r) and alpha (α) values, utilizing **AdamW** and linear decay schedulers to stabilize training under strict memory and parameter constraints.

Skills

- Programming Languages:** C++, Java, Python, TypeScript, JavaScript (ES6), SQL
- Web Technologies:** HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize, Ruby on Rails, Spring Boot, Spring Security, Spring Data, Spring MVC, Spring Cloud, Spring Batch
- Database:** RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
- Cloud:** AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs), GCP
- Tools:** Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Kubernetes, Power BI, QuickSight, JSON, Hibernate
- AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL

Certifications

AWS Cloud Practitioner

Azure Data Fundamentals

Stanford Machine Learning (Coursera)